



RegIntel Copilot

Risk, Fraud & Regulatory Intelligence Copilot
for Banking and NBFC Teams

Hackathon Submission — Project Document

Built entirely with Snowflake CoCo (Cortex Code)

Snowflake Cortex Agent · Cortex Analyst · Cortex Search · Dynamic Tables · Streamlit

October 2026

1. Executive Summary

What it is, what it does, and why it matters.

RegIntel Copilot is an AI-powered compliance assistant built entirely on Snowflake that lets banking compliance teams ask natural language questions and receive governed, explainable, evidence-backed answers — covering the full workflow from fraud signal detection to audit-ready investigation reports.

The core differentiator is not fraud detection. There are thousands of fraud detection systems. The differentiator is: *structured evidence + regulatory knowledge + governed AI + auditability* — in a single conversational interface.

When an analyst asks "*Why is this account high-risk, and what regulatory action is required?*", the system simultaneously queries transaction data for evidence, searches policy documents for applicable rules, and produces a response that cites both — with a clear chain from signal to evidence to finding.

\$210.36M

Modeled Suspicious
Exposure

2,477

Flagged Transactions

4

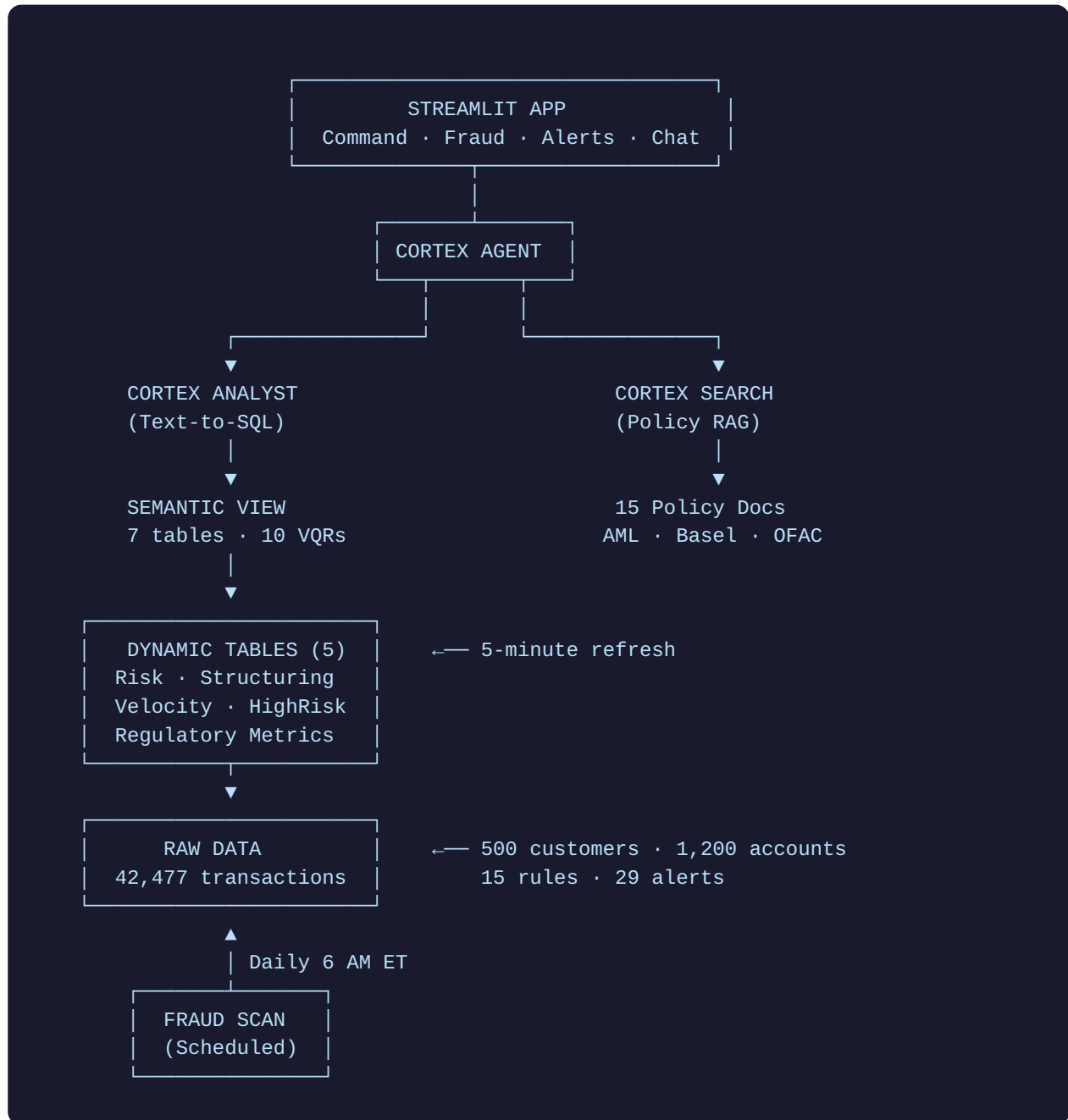
Fraud Typologies

42,477

Total Transactions

2. Architecture

End-to-end Snowflake-native — no external services required.



Snowflake Services Used

Component	Snowflake Service	Purpose
Enrichment Pipeline	Dynamic Tables (5)	

		Near-real-time risk scoring, fraud detection
AI Orchestration	Cortex Agent	Natural language tool routing
Structured Analytics	Cortex Analyst + Semantic View	Governed text-to-SQL with verified queries
Policy Knowledge	Cortex Search	RAG over 15 regulatory documents
Governed Data Layer	Semantic Views	7 tables, 10 VQRs, governed access
Automation	Scheduled Tasks	Daily fraud scan at 6 AM ET
Application	Streamlit in Snowflake	Dashboard, investigation reports, chat
Development	CoCo (Cortex Code)	Entire build lifecycle

3. Data Model & Fraud Patterns

Synthetic Data Layer

Table	Records	Key Fields
CUSTOMERS	500	KYC status, PEP flag, sanctions, risk rating, occupation, income
ACCOUNTS	1,200	5 types (savings, checking, business, loan, FD), status, balance
TRANSACTIONS	42,477	8 types, 4 currencies, channels, risk scores, fraud labels
REGULATORY_RULES	15	BSA/AML, Basel III, OFAC, FATF with thresholds
ALERTS	29	5 statuses, severity, assigned investigator, SAR filing
POLICY_DOCUMENTS	15	AML, Basel III, OFAC, KYC, filing templates

Embedded Fraud Patterns (Labeled Ground Truth)

Pattern	Transactions	Exposure	Detection Method
STRUCTURING	1,357	\$12.18M	Repeated sub-threshold deposits within aggregation windows
LAYERING	200	\$55.16M	Rapid multi-hop transfers through account chains
ROUND-TRIP	120	\$53.23M	Funds returning to origin via intermediaries
RAPID MOVEMENT	800	\$89.79M	Burst high-value transfers inconsistent with baseline
TOTAL	2,477	\$210.36M	

Enrichment Pipeline (5 Dynamic Tables)

Dynamic Table	Rows	What It Computes
CUSTOMER_RISK_PROFILE	500	

		Composite 0–100 risk score (KYC + PEP + sanctions + behavior)
STRUCTURING_DETECTION	296	Windowed analysis: sub-threshold cash deposits in 48h windows
TRANSACTION_VELOCITY	33,120	Daily volume vs 30-day rolling average, 3x+ anomaly flags
HIGH_RISK_TRANSACTIONS	2,494	Multi-signal scoring: txn risk + customer + jurisdiction + pattern
REGULATORY_METRICS	1	23 compliance KPIs refreshed every 5 minutes

4. Application Walkthrough

The application has five views, designed around the compliance analyst's workflow.

4.1 Command Center

Hero KPIs — How much? How many? What needs attention?

\$210.36M

Suspicious Exposure

2,477

Flagged Txns

8

Active Alerts

2

SLA At Risk

Below the hero KPIs: Customer Risk Distribution chart (Critical/High/Medium/Low), Fraud Pattern Activity chart (by typology), and an **Investigation Queue** showing the highest-priority cases sorted by severity and exposure.

4.2 Fraud Detection

Two tabs: Structuring Detection + High-Risk Transactions

Structuring Detection: Shows 296 findings — repeated sub-threshold cash deposits within 48-hour aggregation windows, classified by severity (CRITICAL, HIGH, MEDIUM, LOW). Filterable by severity level.

High-Risk Transactions: 2,494 transactions scored across multiple risk signals. Filterable by fraud pattern type. Shows transaction ID, account, customer name, amount, risk score, and primary risk reason.

4.3 Alerts & Investigation

Alert Queue + Investigation Report Generator

Alert Queue: All 29 alerts filterable by status (Open, Under Investigation, Escalated, Closed). Joined with regulatory rules to show which regulation triggered each alert.

Investigation Report Generator: Select any alert → click "Generate Investigation Report" → the system produces a structured 7-section report:

1. **Subject Information** — Customer name, ID, type, nationality, account details
2. **Risk Assessment** — Composite score, PEP/sanctions flags, transaction counts
3. **Alert Details & Fraud Typology** — Triggered rule, threshold, description
4. **Supporting Transaction Evidence** — Individual transactions with amounts and dates
5. **Applicable Regulatory Requirements** — Retrieved from policy docs by Cortex Agent
6. **Investigation Findings** — Current status, SAR filing status, resolution notes
7. **Recommended Next Steps** — Context-aware (PEP → senior approval, expired KYC → refresh, etc.)

Includes disclaimer: "This report is generated by an AI system and requires review by a qualified compliance analyst before any regulatory action is taken."

4.4 Customer Risk

Customer profiles with composite risk scoring

500 customers filterable by risk rating and minimum composite score. Shows PEP/sanctions flags, account count, balance, flagged transaction count, and active alerts. Includes KYC Renewal Required section for expired/failed/pending verifications.

4.5 Ask Copilot

Natural language interface backed by Cortex Agent

Chat interface powered by the REGINTEL_COPILOT_AGENT. The agent routes questions to the appropriate tool:

- **Data questions** → Cortex Analyst queries the semantic view (governed SQL)
- **Policy questions** → Cortex Search retrieves regulatory documents
- **Combined questions** → Both tools used; response cites evidence + regulation

Example: *"Does the structuring pattern on ACCT-00000001 meet our SAR investigation criteria? Cite the relevant policy."* → Returns transaction evidence + BSA/SAR policy section + 30/60-day filing timeline.

5. Hero Workflow: Signal → Evidence → Finding

This is the core demo: the complete compliance workflow in one interaction.

The flow: Detection → Investigation → Regulatory Reasoning → Evidence → Documentation

Step 1: Detection

The Command Center shows **\$210.36M** in modeled suspicious exposure across **2,477** flagged transactions. The Investigation Queue surfaces the highest-priority case: a **CRITICAL** structuring alert on account ACCT-0000001.

Step 2: Investigation

The analyst clicks into Alerts & Investigation → selects the alert → clicks **"Generate Investigation Report."** The system compiles: customer profile (KYC status, PEP flag, risk score), account details, the specific transactions that form the structuring pattern, and the regulatory rule that triggered the alert.

Step 3: Regulatory Reasoning

The analyst switches to Ask Copilot and asks: *"Does this structuring pattern meet our SAR investigation criteria? Cite the relevant policy."*

The Cortex Agent responds with a structured assessment:

- **Assessment:** Potentially reportable — analyst review required
- **Transaction Evidence:** 17 cash deposits, \$164,700 aggregate, 48-hour window
- **Risk Indicators:** Repeated sub-threshold deposits, aggregated cash activity, pattern inconsistent with baseline
- **Regulatory Context:** BSA/SAR policy, Section 4.2
- **Filing Timeline:** 30 calendar days from initial detection; up to 60 days if no suspect identified (per FinCEN)
- **Disclaimer:** AI-generated analysis; final determination requires authorized compliance review

Step 4: Evidence & Documentation

The investigation report contains all evidence, regulatory context, and recommended next steps — ready for compliance review and case management filing.

6. Regulatory Scope

Framework	Coverage	Key Thresholds
BSA/AML	CTR reporting, SAR filing, structuring detection, wire record-keeping	\$10,000 CTR; SAR per 31 CFR §1020.320
Basel III	Liquidity Coverage Ratio, Capital Adequacy Ratios	100% LCR minimum; 4.5% CET1, 8% total capital
OFAC	SDN list screening, blocking and reporting	All transactions screened; 10-day blocking report
FATF	High-risk jurisdiction guidance, EDD recommendations	Jurisdiction lists per current FATF publications

SAR Filing Nuance: The prototype models the \$5,000 threshold applicable to banks under 31 CFR §1020.320. Thresholds vary by institution type and jurisdiction. The 30-day filing deadline includes the FinCEN extension: if no suspect is identified, filing may be delayed up to 60 calendar days total.

Production note: In production, regulatory scope would be scoped to the specific institution's obligations. FATF high-risk jurisdiction lists are time-sensitive and must be updated per current FATF publications. This prototype includes all frameworks to demonstrate cross-domain regulatory reasoning.

7. CoCo Usage Across the Full Lifecycle

All hackathon solutions must use CoCo across the full lifecycle. Here is the evidence for each phase.

Phase	What CoCo Did	Evidence
Planning	Designed data model, architecture, regulatory scope	Schema design, table structures, fraud pattern selection
Data Generation	42,477 synthetic transactions with 4 labeled fraud patterns	Referentially consistent FKs, realistic distributions
Pipeline	5 dynamic tables with windowed aggregations, composite scoring	CREATE DYNAMIC TABLE statements, 5-min refresh
Semantic Layer	7-table semantic view with 10 verified queries	sv-generate, sv-deploy via CoCo CLI
Search Service	Cortex Search over 15 policy documents	CREATE CORTEX SEARCH SERVICE
Agent	Dual-tool Cortex Agent (Analyst + Search)	agent-write, agent-deploy via CoCo CLI
Application	5-view Streamlit app with investigation reports	627-line streamlit_app.py
Automation	Scheduled daily fraud scan task	CREATE TASK with CRON schedule
Testing	Validated agent responses, verified SQL accuracy	DATA_AGENT_RUN test queries, VQR validation

CoCo Capabilities Demonstrated

- Synthetic data generation with embedded fraud signals
- Data pipeline creation using dynamic tables
- Semantic model and ontology authoring with verified queries
- Document processing and indexing via Cortex Search
- Streamlit report and app generation
- Scheduled automation via Tasks
- Guardrails through semantic view scoping (governed answers only)

8. Limitations & Future Directions

Limitations

This is a **production-oriented prototype** built on synthetic data in a hackathon environment. A production deployment would require:

- Real transaction data with proper access controls and encryption
- Model validation and explainability testing
- Regulatory change management processes
- Disaster recovery and audit logging
- Security review and penetration testing
- Integration with existing case management and filing systems (e.g., FinCEN BSA E-Filing)

Future Directions

Enhancement	Description
Network Graph Analysis	Counterparty relationship mapping for layering chain visualization
MCP Integration (Slack)	Real-time alert notifications to compliance Slack channels
SAR Auto-Filing	Generate FinCEN-ready SAR XML from investigation reports
ML Anomaly Detection	Train models on labeled fraud data for beyond-rule-based detection
Org Rollup	Multi-entity compliance views for holding companies
Reusable Skill	Publish as Cortex Code skill for other teams